EXPERIMENT 5b – DATA VISUALIZATION USING MATPLOTLIB

Aim

To analyze and visualize the tourism dataset using different graphical representations.

Procedure

1. Import the required libraries and load the tourism dataset.
2. Analyze the important columns in the dataset.
3. Create different graphs using tourism data.
4. Visualize Visitors, Rating, Revenue, Country, and Category.
5. Analyze the results obtained from the visualizations.

Program

1. **Import Libraries and Read Dataset**

```
import pandas as pd
import matplotlib.pyplot as plt

# Read dataset
df = pd.read_csv(r"C:\Users\kalap\Downloads\tourism_dataset (1).csv")
# Display first five records
    display(df.head())
```

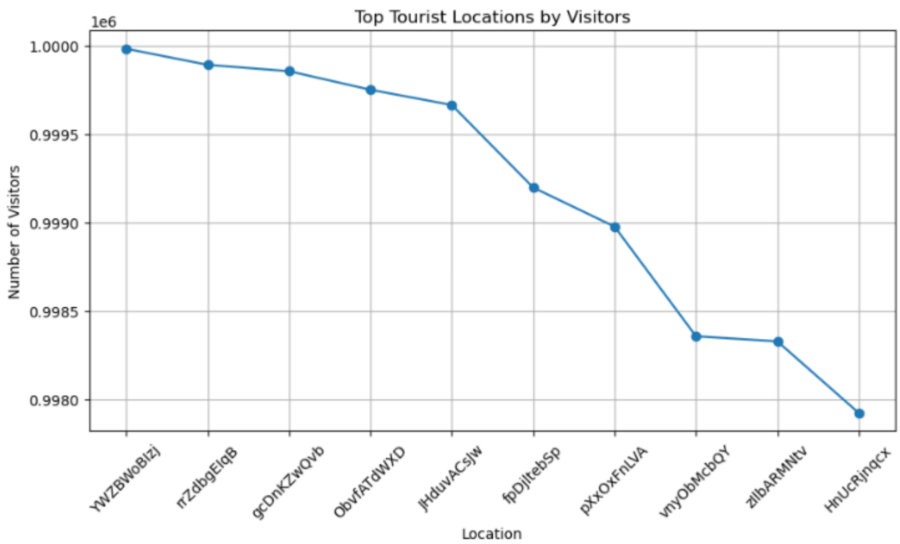
```
display(df.head())
```

	Location	Country	Category	Visitors	Rating	Revenue	Accommodation_Available
0	kuBZRkVsAR	India	Nature	948853	1.32	84388.38	Yes
1	aHKUXhjzTo	USA	Historical	813627	2.01	802625.60	No
2	dIrdYtJFTA	Brazil	Nature	508673	1.42	338777.11	Yes
3	DxmlzdGkHK	Brazil	Historical	623329	1.09	295183.60	Yes
4	WJCCQlepnz	France	Cultural	124867	1.43	547893.24	No

2. **Program 1: Line Chart – Tourist Visitors**

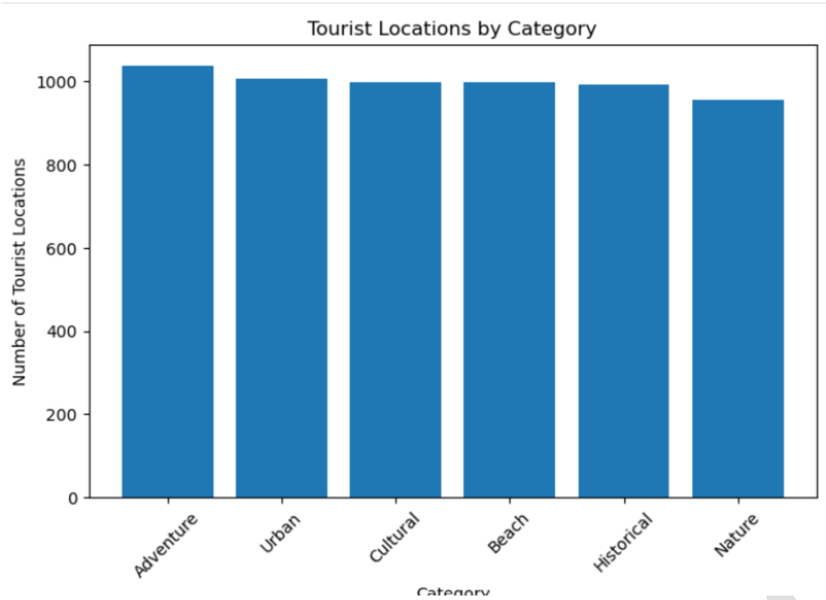
```
# Top 10 locations for better visualization

top10 = df.sort_values("Visitors", ascending=False).head(10)
```

```
plt.figure(figsize=(10,5))

plt.plot(top10["Location"], top10["Visitors"], marker="o")

plt.title("Top 10 Tourist Locations by Visitors")
plt.xlabel("Location")
plt.ylabel("Visitors")

plt.xticks(rotation=45)

    plt.show()
```



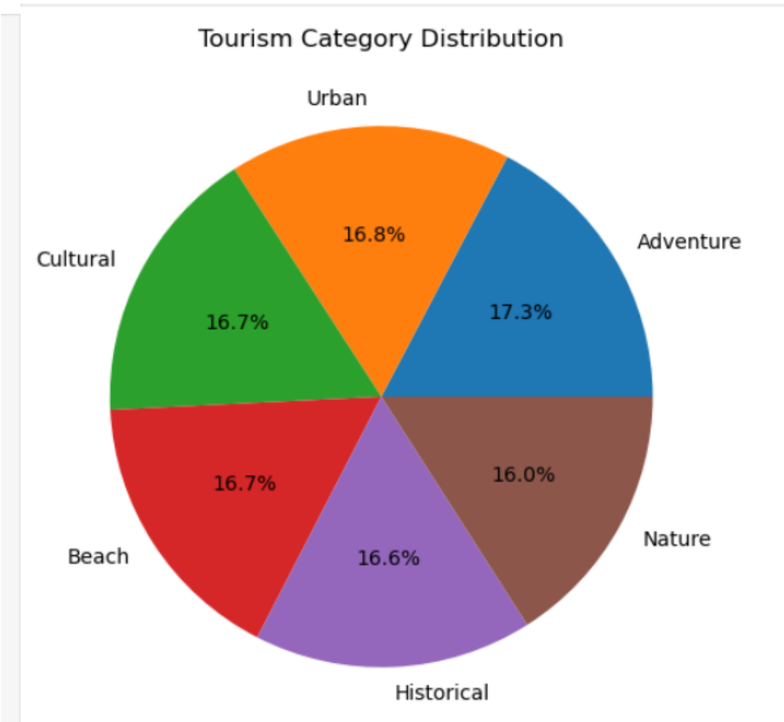
3. Program 2: Bar Chart – Tourist Visitors

```
top10 = df.sort_values("Visitors", ascending=False).head(10)

plt.figure(figsize=(10,5))

plt.bar(top10["Location"], top10["Visitors"])

plt.title("Top 10 Tourist Locations by Visitors")
plt.xlabel("Location")
plt.ylabel("Visitors")

plt.xticks(rotation=45)

    plt.show()
```



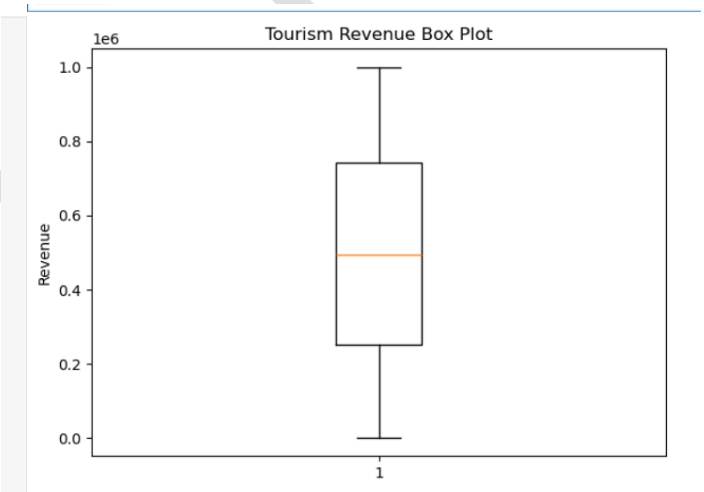
4. **Program 3: Pie Chart – Tourism Categories**

```
category_count = df["Category"].value_counts().head(5)

plt.figure(figsize=(8,8))

plt.pie(
    category_count,
    labels=category_count.index,
    autopct="%1.1f%%"
)

plt.title("Tourism Category Distribution")

plt.show()
```



5.Box Plot

```
plt.figure(figsize=(7,5))

plt.boxplot(df["Revenue"].dropna())

plt.title("Tourism Revenue Box Plot")
plt.ylabel("Revenue")
plt.show()
```



RESULT: